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Summary of

End of studies project

Mechanical Engineering option Industrial Energy carried out at ENSAIS

Development of a test bench and characterization of the GEET process by P. Pantone for reforming hydrocarbons.

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A) Introduction: the current disturbing state of the planet.

The current climatic upheaval of the planet is very likely to come from human activity and the industrial and economic development of the last 200 years. This development is generally accompanied by the combustion of fossil fuels which release large quantities of polluting gases into the atmosphere. *(Fig* 1) Fossil fuels, by their name, constitute the memory of

the Earth and result from the natural absorption of carbon by the earth. Their combustion upsets in a dazzling way this natural carbon cycle in place for millions of years with serious consequences for the global balance of the planet.

In effect ; carbon dioxide, which originally allowed the life to develop on Earth, is today the main greenhouse gas responsible for climate change. *(Fig 2 and* 3) If it is not dangerous for humans, it is not the same for the other gases resulting from "direct" combustion

of hydrocarbons. **P.Pantone's GEET (Global Enviroment Energy Technology) process makes it possible, through a pre-treatment of conversion of the hydrocarbon mixture before combustion, to greatly reduce emissions of carbon pollutants. In this sense, it constitutes one of the first steps towards "hydrogen generation".** The aim of this project is the characterization of the process by developing a test bench.

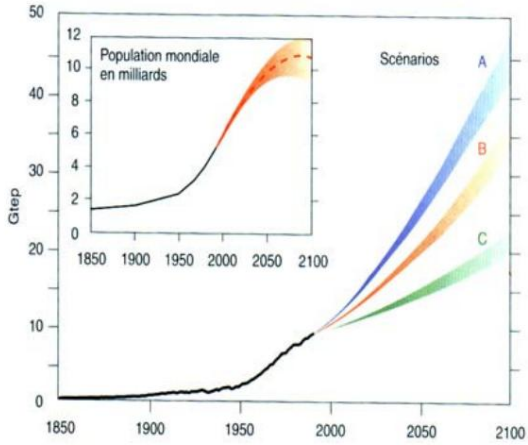

Fig 1: Global primary energy consumption in Gtoe. History since 1850 and forecast according to 3 scenarios. .The box shows the total world population.

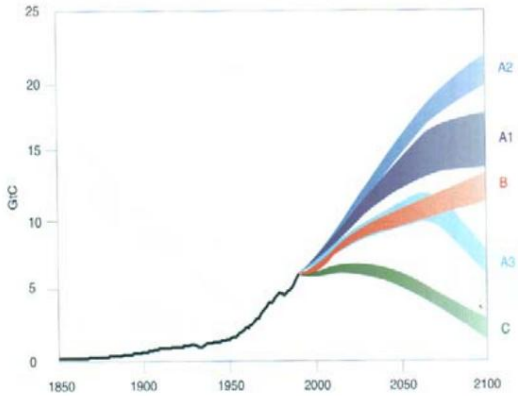

Fig 2: Evolution of the quantity of CO2 in the atmosphere for 150 years and evolution according to 5 more or less ecological scenarios

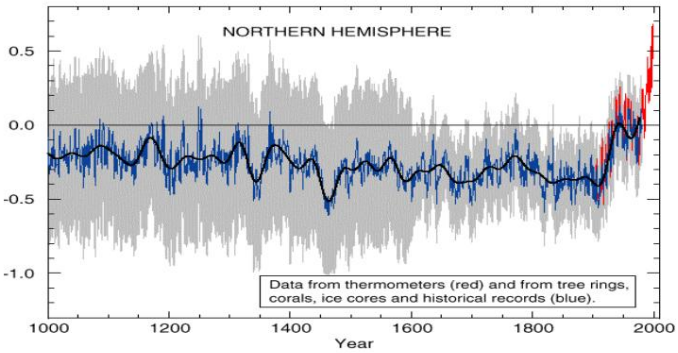

Fig 3: Evolution of the temperature since the year 1000 ***B) Presentation of the GEET process by P.Pantone.***

1) Principle

Paul Pantone is an American electrical engineer who converted to energy after carrying out fuel treatment experiments that led to the following assembly (patent reference: WO 96/14501 and PCT/US95/14383). **The evaporator-bubbler contains a mixture of water and gasoline.**

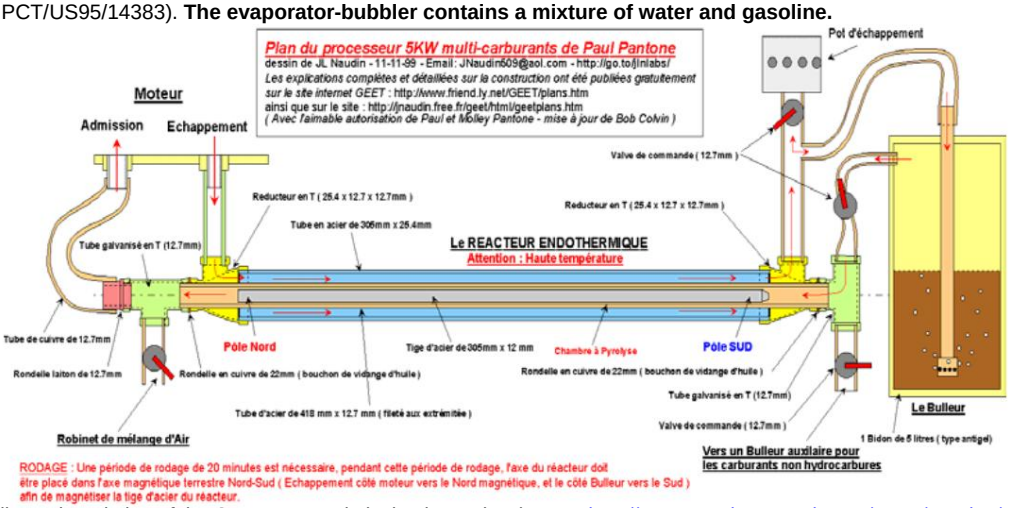

Fig 4: Free dimensioned plan of the GEET system in its basic version (source http:// www.exoticresearch.com/ store/ catalog/ geet/ geet.htm)

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The principle is simple: the heat, normally lost, of the exhaust gases is recovered to treat the intake gases (resulting from the volatilization of fuel and water in the evaporator) in the reactor by an electro- magnetic-chemical, it is complex. The heated admission gases circulate in an annular space created by the geometry of the reactor and are subjected to an oriented magnetic field NS.
The magnetic field observable during operation reflects the presence of ionized species and/or moving electric charges. This field magnetizes all parts of the reactor after a certain period of operation. The following diagram shows the process flows.

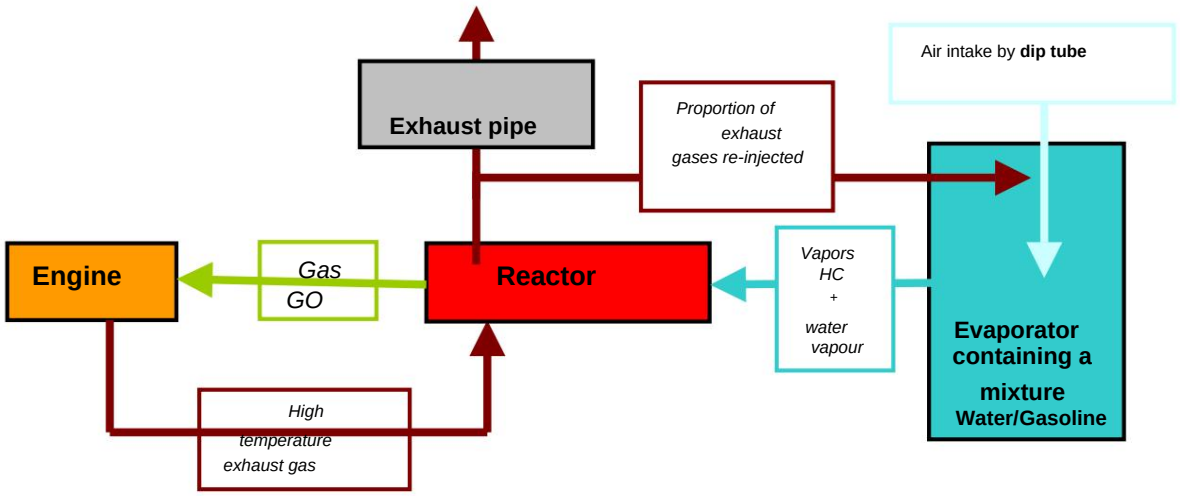


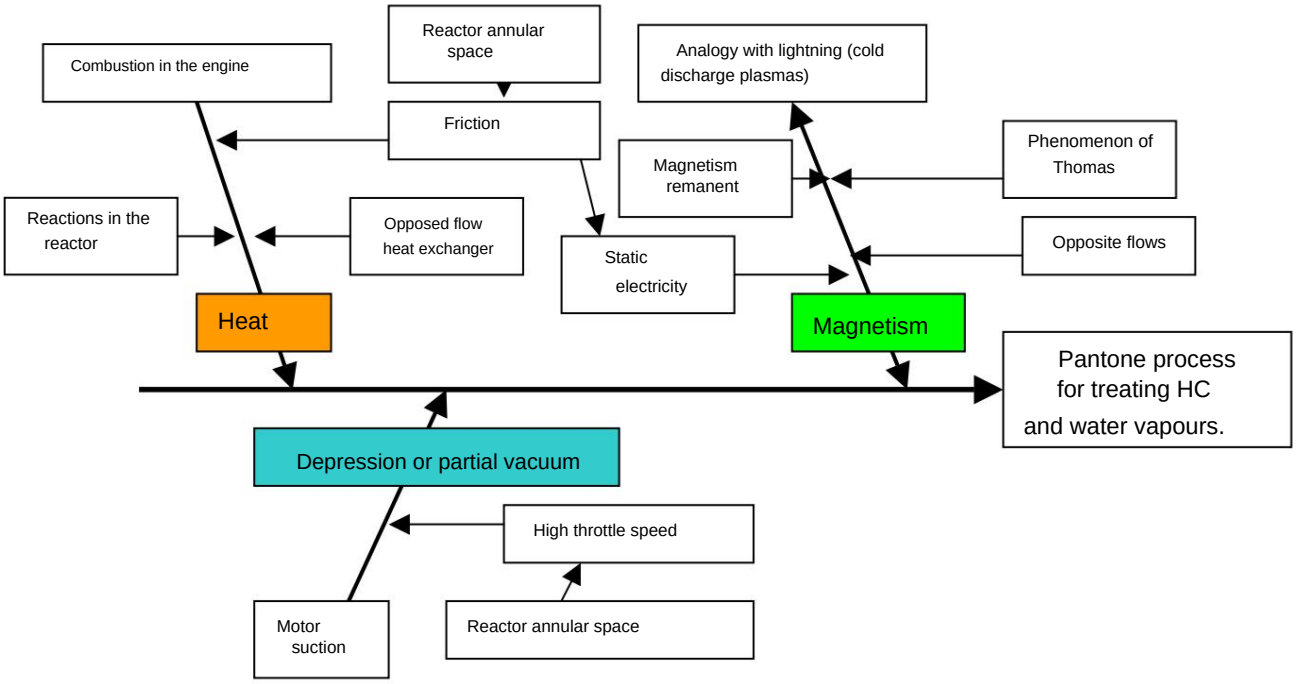
Fig 5: Flow diagram of the process.

The advantage of the process is its relative simplicity of assembly (stacking of tubes) and its adaptation to an already existing engine, the costs of depollution, use and maintenance are therefore greatly reduced in comparison with other depollution processes (fuel cell, solar). In this sense, this process constitutes a chance for the depollution of the great emerging demographic powers which will double the world energy consumption in the next 30 years.

2) Causes

The Pantone process is defined, by its inventor, as a process for treating fuel by plasma discharge (lightning) but it is, for the moment, difficult to explain scientifically the causes of its reaction. Nevertheless, 3 fundamental causes necessary for the reaction have been determined, they are synthesized in the following diagram.

Fig 6: Fishbone cause-effect diagram.



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Heat, magnetism (cause or consequence) and depression are the 3 primary causes of the process, according to Pantone, there would be 38 distinct phenomena, causes or effects involved during the reaction. Additional specialized research in various fields is therefore desirable in order to fully understand the process.

C) Design and development of a test bench

To characterize the process, a test bench based on a 4 kW generating set was entirely built. A circuit with 2 evaporators has been created in order to be able to constantly monitor the Water/Hc flow rate ratio. It comprises sensors measuring the essential quantities in the characterization of a carburization process as shown in the following documents.

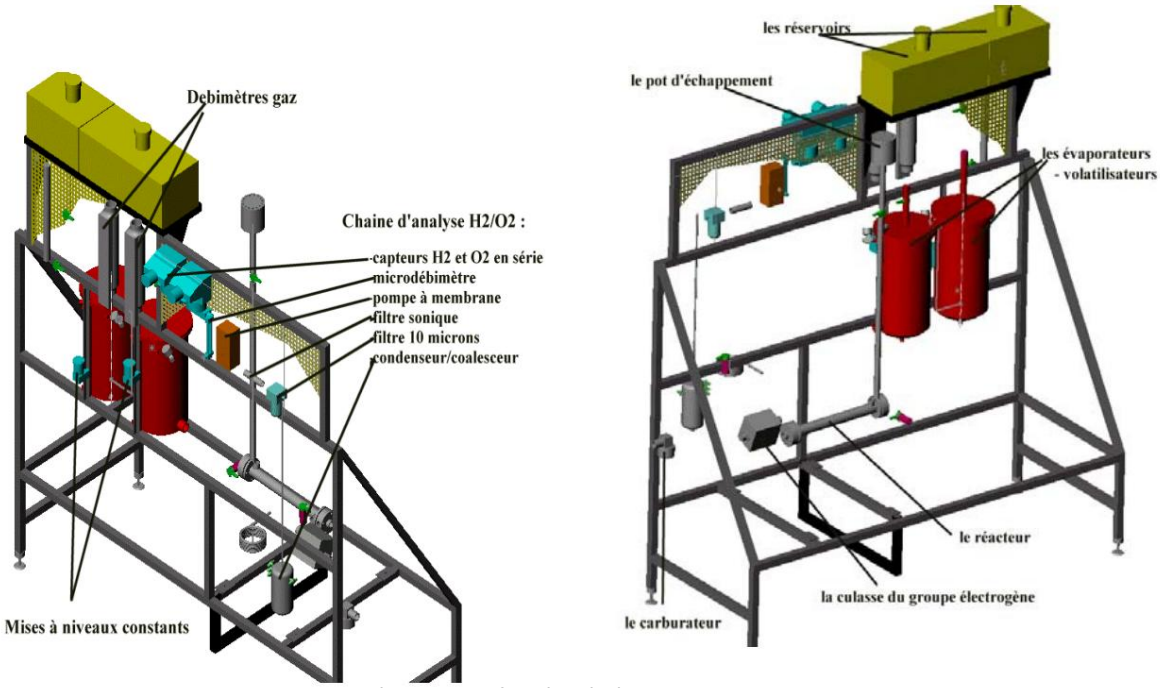

Fig 7: Layout of test bench elements

Fig 8: Photograph of the complete bench

Fig 9: Detail photograph of the gas flowmeters and vapor outlet from the evaporators.

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D) Results in consumption and depollution.

1) Consumption

The consumption on gasoline in the Pantone process remains almost identical to the carburettor assembly of origin, it shows a slight decrease of 5% maximum. The assembly carried out is a basic non-optimized assembly with a very summary control of the quantity of fuel injected, the development of an improved assembly could highlight a more significant drop in consumption. Water consumption is approximately equal to gasoline consumption but is condensable in the exhaust circuit. Anyway, improved assembly studies are necessary if we want to expect the same performance as the latest injection systems in the industry.

2) Depollution

The process, well regulated, presents a remarkable depollution. This reflects the conversion of gasoline vapors into a gas (GEET gas) whose combustion is much more complete and "clean" than that of gasoline vapors.

Fig. 10: Pollutants during carburettor operation in function of the electric power supplied by the group generator

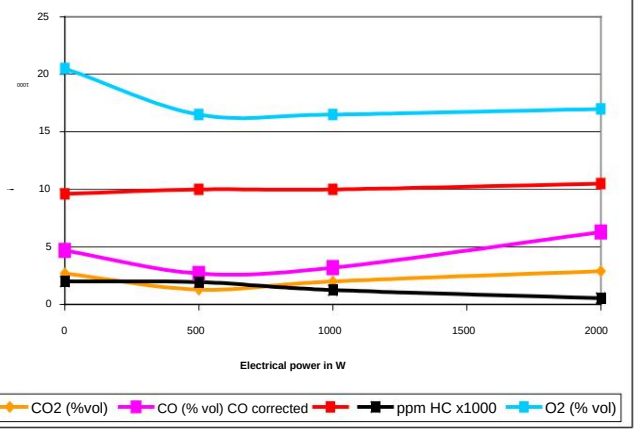


Fig. 11: Pollutants in Patnone operation according to the electrical power supplied by the generating set

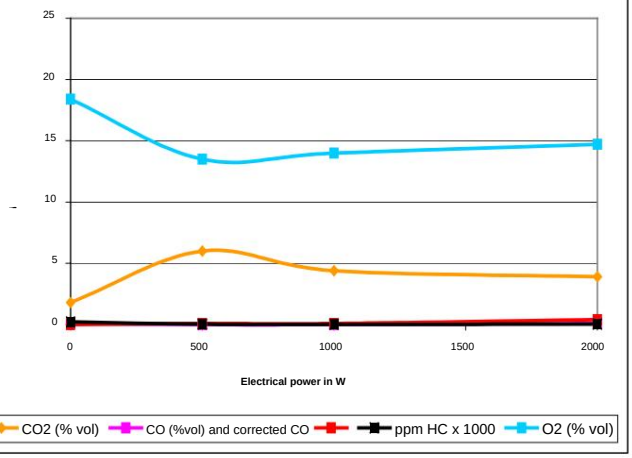
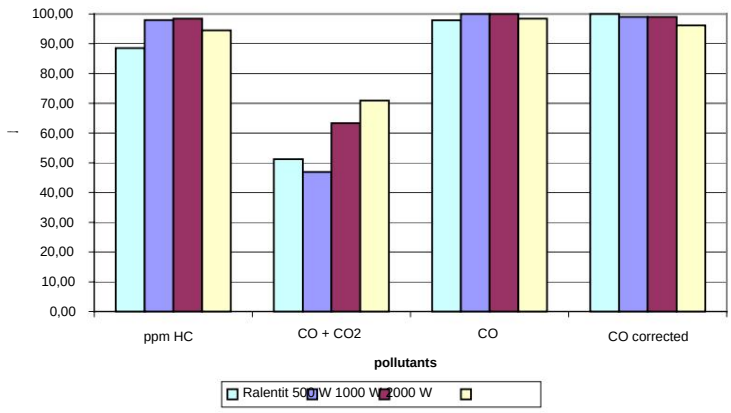


Fig. 12: Decrease in process pollutants compared to carburettor operation depending on the electrical power of the generator set



The depollution (of the pollutants measured) is remarkable: in fact, only CO2 remains (6% vol. maximum) in a proportion approximately less than half of the proportion theoretically expected in the case of gasoline combustion (about 14% vol). Once the process is set, the exhaust gases are respirables.

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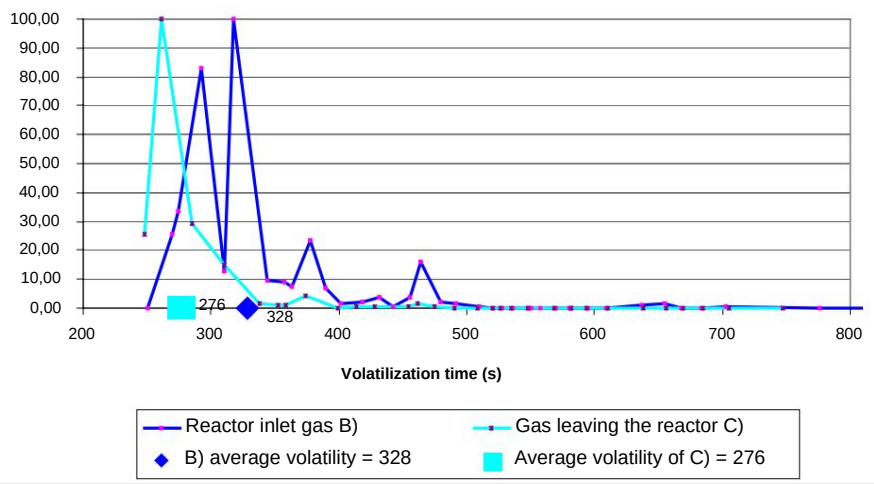
Remarks :

- the depollution measured was carried out at steady state and at the point of best depollution after adjustment of the various dosing and richness valves. Depollution at variable speed is more complex and requires the automation of the different flow rates, this would also make it possible to obtain a single power control.
- It is not excluded that the process rejects other unmeasured pollutants, such as Nox, but the recirculation of exhaust gases makes it possible to greatly reduce the Nox created at high temperature (EGR, Exhaust Gas Recirculation, is currently used by car manufacturers to reduce NOx emissions)

E) Gas analysis results

A gaseous analysis by chromatography was carried out, it made it possible to prove the conversion of the hydrocarbon vapors into a single more volatile compound.

Fig 13: Chromatograms of the gas entering and leaving the reactor



Reading a chromatogram:
- each peak corresponds to a compound the height of each peak gives its mass proportion in the mixture
- its location on the time scale reflects its volatility.

GEET gas is therefore a simpler and more volatile gas than petrol vapours, which would explain the strong depollution. But other gaseous analyzes (catharometry specific to reforming) showed that this volatile compound was neither hydrogen nor methane. But the reliability of the analyses, particularly in hydrogen needs to be reviewed. Indeed, there are strong risks of diffusion of hydrogen before the analysis devices. Additional work will be necessary to precisely qualify this gas.

F) Perspectives.

This study showed the depollution efficiency of the process. The process therefore deserves additional studies in order to achieve a more complete characterization of the conversion process. in parallel adaptability studies on other systems using hydrocarbons will have to be carried out.

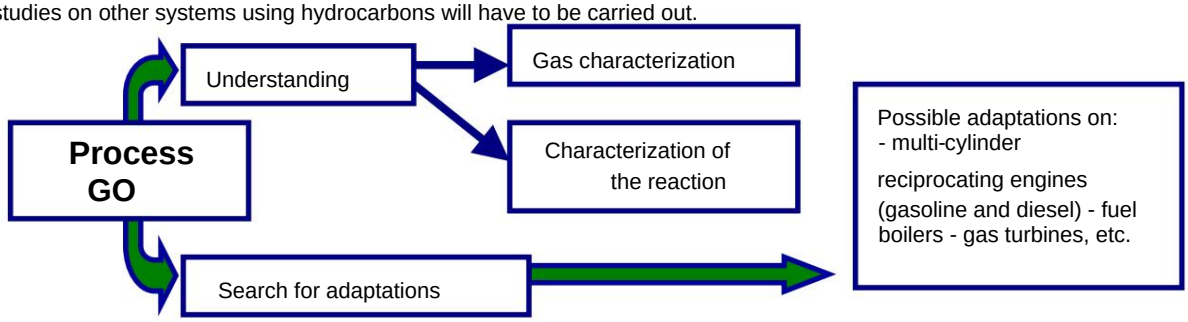


Fig 14: Perspectives of the GEET process

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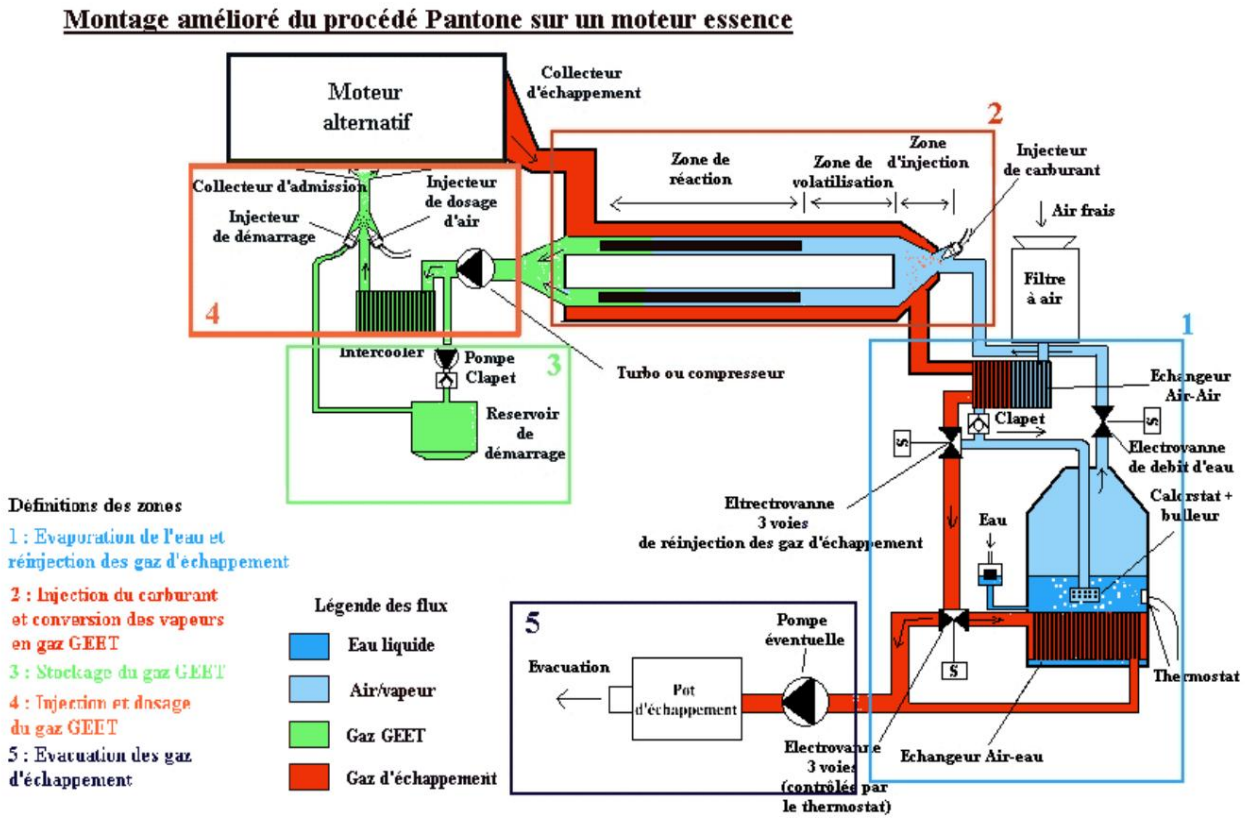
The assembly, experimental, that we made was basic and had 3 major drawbacks:

- decrease in the rate of filling (30 to 50% by weight) and engine emptying due to pressure drops in the intake and exhaust circuit
- poor control of the fuel supply (evaporator and authority of the valves) - unadjusted ignition advance (there is a good chance that the more volatile GEET gas will burn faster)

So, taking into account these 3 defects and other findings and experimental observations, here is the improved mounting that we recommend Fig. 15:



This assembly is more complex than the assembly carried out but will undoubtedly allow a better control of the power (providing the flexibility of use of the traditional power supply systems) and of the depollution. It will also perhaps allow us to see an improvement in consumption…but it remains to be designed & tested on a bench.

G) Other observed phenomena/characteristics.

Here are some other features I noticed during my experiments. Each of these points deserves further experimentation which could eventually lead to industrial exploitation.

1) reduction of consumption of 20 to 50% on Diesel engine. Many testimonies (in particular agricultural tractors) show a gain in consumption of at least 20% observed after the modification of their engine and this under the same conditions of use. (see the web page on the site www.quanthomme.com concerning tractors modified by doping with water, assembly number 22). **2) pollution reduction of up to 95%** depending on the pollutants measured on the prototype of l'ENSAIS.

3) possible loop operation: the system allows a reinjection of part of the exhaust gases but under certain conditions it was possible for us to

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reinject 100% of the exhaust gases. The engine then ran for about 20 minutes. That is to say well beyond the purging of the various circuits.

4) limits of rotation regimes pushed back. We noticed a very low idle: 300 rpm on a 4-stroke 300cm3 engine, this is to be compared to the 1200 rpm of the original carburettor operation. On the other hand, the maximum rotation speed (unladen) is increased by approximately 30%.

5) the spark plug intact after 1 hour of operation. A new plug fitted when the process is hot remains golden after one hour of operation.

6) exhaust temperature increased by about 40%. Exhaust pipe fitted to red. However, we did not notice any deterioration on the engine (seats & valves intact).
Nevertheless, it would be necessary to do fatigue tests to confirm this behavior. (Our engine only ran for about 50 hours). This excess temperature proves the combustion of a compound different from pure gasoline.

These points are only the most interesting findings but each requires investigation. complementary. Some could quickly lead to industrial exploitation.

G) Conclusion: a promising process but additional studies needed

We are in the presence of a promising process on several points which deserves additional studies of characterizations and industrialization. Once characterized and improved, the process, developed industrially and distributed on a large scale, could lead to a spectacular drop in atmospheric pollution, especially in emerging demographic powers that would have access to cheaper and decentralized energy. Indeed, for reasons of cost and quantity, they cannot reasonably use the latest energy technologies (fuel cell, nuclear, etc.), moreover they generally have significant fossil energy resources.